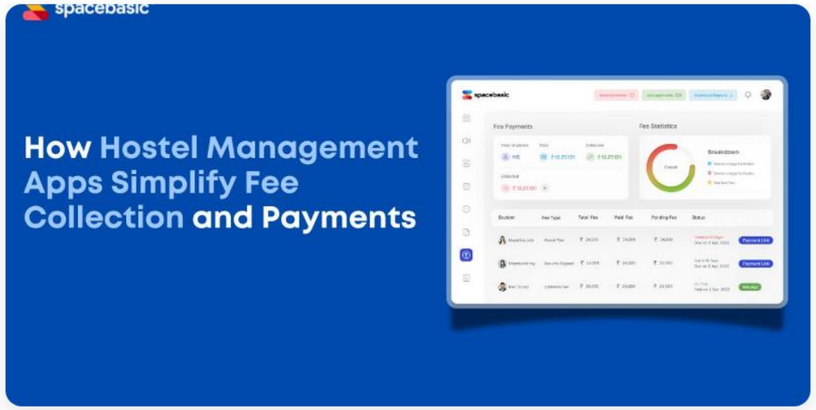
Hostel Management System

Object-Oriented Programming Lab (C++)

Iqra University

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Instructor: Sir Khawaja Badar



Project Overview

A comprehensive digital solution for institutional accommodation.

Introduction & Challenges

Digital Evolution

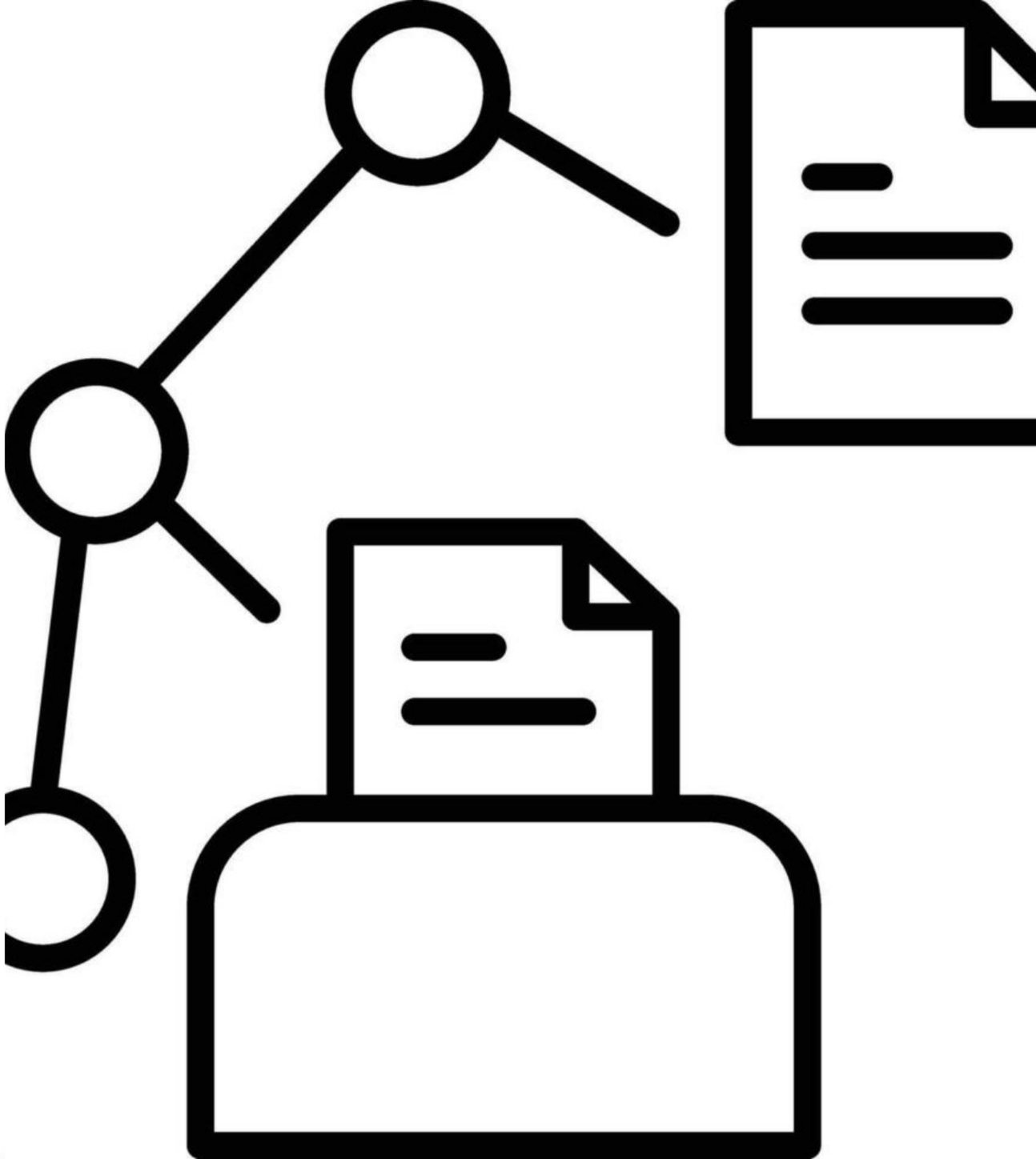
Developing a robust hostel management system using C++ and Object-Oriented Programming (OOP) to modernize administrative tasks.

Existing Problems

- Data loss and duplication
- Difficult room allocation
- Slow fee management
- Excessive paperwork

Core Objectives

- Centralized student records
- Automated room allocation
- Digital Fee management
- Full CRUD operations
- Paperless administration



Strategic Benefits

100%

PAPERLESS & EFFICIENT

Enhanced Efficiency

By digitizing records, we achieve unparalleled accuracy and security. The system ensures that administrative overhead is minimized while data integrity is maximized.

- Accuracy
- Security
- Speed

System Features & Modules

Student Registration

Digital onboarding and profile management.

Room Management

Dynamic allocation and occupancy tracking.

Fee Management

Transaction recording and automated history.

Advanced Maintenance

Search & Control

The system provides a robust search inquiry interface for instant data retrieval. Administrators can perform record modifications and deletions with strict data validation protocols.

- Instant student lookup
- Dynamic record updating
- Secure deletion safeguards



Technical Architecture

Leveraging Object-Oriented Programming (OOP) Principles.

Core OOP Concepts Used

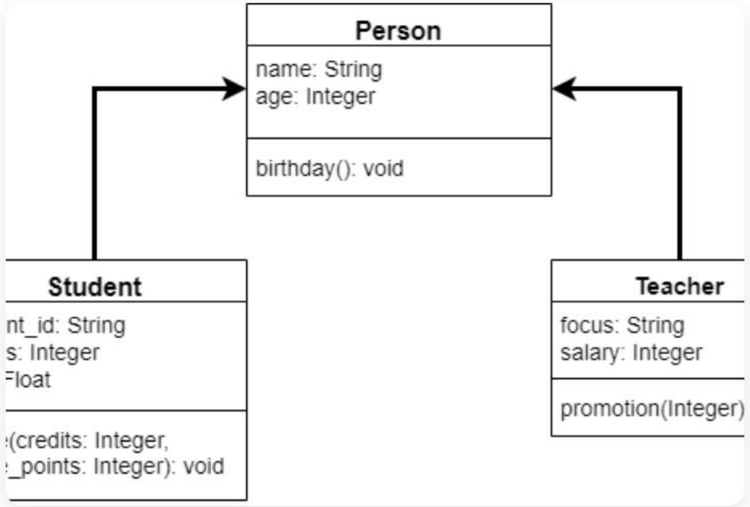
Classes & Objects
Used as blueprints for real-world entities like Students and Rooms.

Encapsulation
Ensuring data security by bundling data and methods within objects.

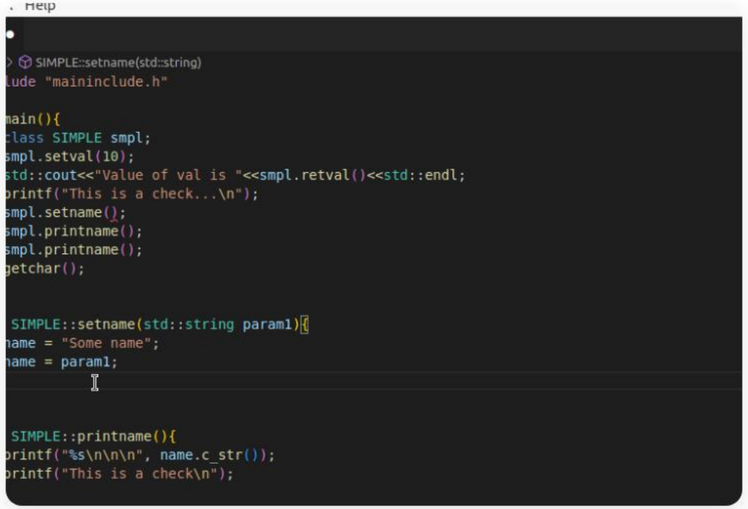
Inheritance
Logical class hierarchy (Person → Student) for code reusability.

Polymorphism
Flexibility through method overriding and overloading.

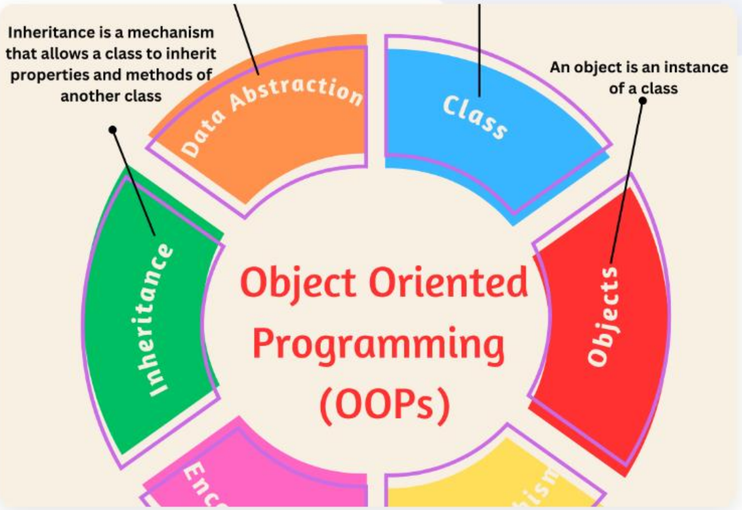
Implementation Visuals



Class Inheritance



File persistence



Modular Design

Operational Workflow

Launch

Initiate Admin App

Manage

Input Student Data

Allocate

Assign Hostel Rooms

Record

Log Fee Payments

Report

Final Summary



Conclusion

Efficient digital hostel administration achieved.

Any Questions?

Iqra University - CS Department

Image Sources



https://cdn.prod.website-files.com/647dcc7d54b50a794ff371f6/6870b9658eb5cb733d3d0548_How%20Hostel%20Management%20Apps%20Simplify%20Fee%20Collection%20and%20Payments.jpg

Source: www.spacebasic.com



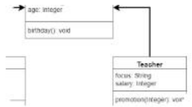
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